Vive Tracker Link Monitor — User Guide

A field tool for checking the health of HTC Vive tracker links during a session. It answers one question while your trackers are running: **is anything wrong with the radio (dongle) or optical (lighthouse) link right now, and which dongle is to blame?** — and saves a report you can compare run to run.

> This is a self-contained guide intended to be handed out. It covers running a session, reading the results, and the files the tool produces.

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1. How it works (in one paragraph)

The tool attaches to **SteamVR** (via OpenVR) and polls every tracked device ~250 times a second. For each tracker it measures two **independent** failure modes and never lets them overlap:

- **Radio link** — the wireless link between a tracker and its **USB dongle**. Counted as lost when the device reports disconnected, or when its input packet counter stalls. A radio problem points at the **dongle** or its placement.
- **Lighthouse (optical)** — line of sight between a tracker and the **base stations**. Counted (only while the radio link is up) when the tracker is connected but out of range. An optical problem points at **occlusion or range** to the base stations, not the dongle.

Each tracker is mapped to the dongle it is paired with, so results can be aggregated **per dongle** — which is what makes a single bad dongle obvious.

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2. Requirements

- A **Windows PC** with **SteamVR installed**.
- **SteamVR running and active** — a headset detected, or a Null/headless driver enabled. The app talks to SteamVR, not to the trackers directly.
- The **Vive trackers powered on** (the trackers themselves, not just the headset) with their **USB dongles plugged in**.
- No network connection, accounts, or extra drivers are required.

> **If Windows blocks the download** (SmartScreen / Smart App Control): the `.exe` is unsigned. Right-click it → **Properties** → tick **Unblock** → **OK**, or temporarily turn Smart App Control off, then launch it again.

When the app opens, the **title bar** ends with `build <date>` (for example `build 2026-06-10h`). Quote that build when reporting anything, so it's clear which version produced a result.

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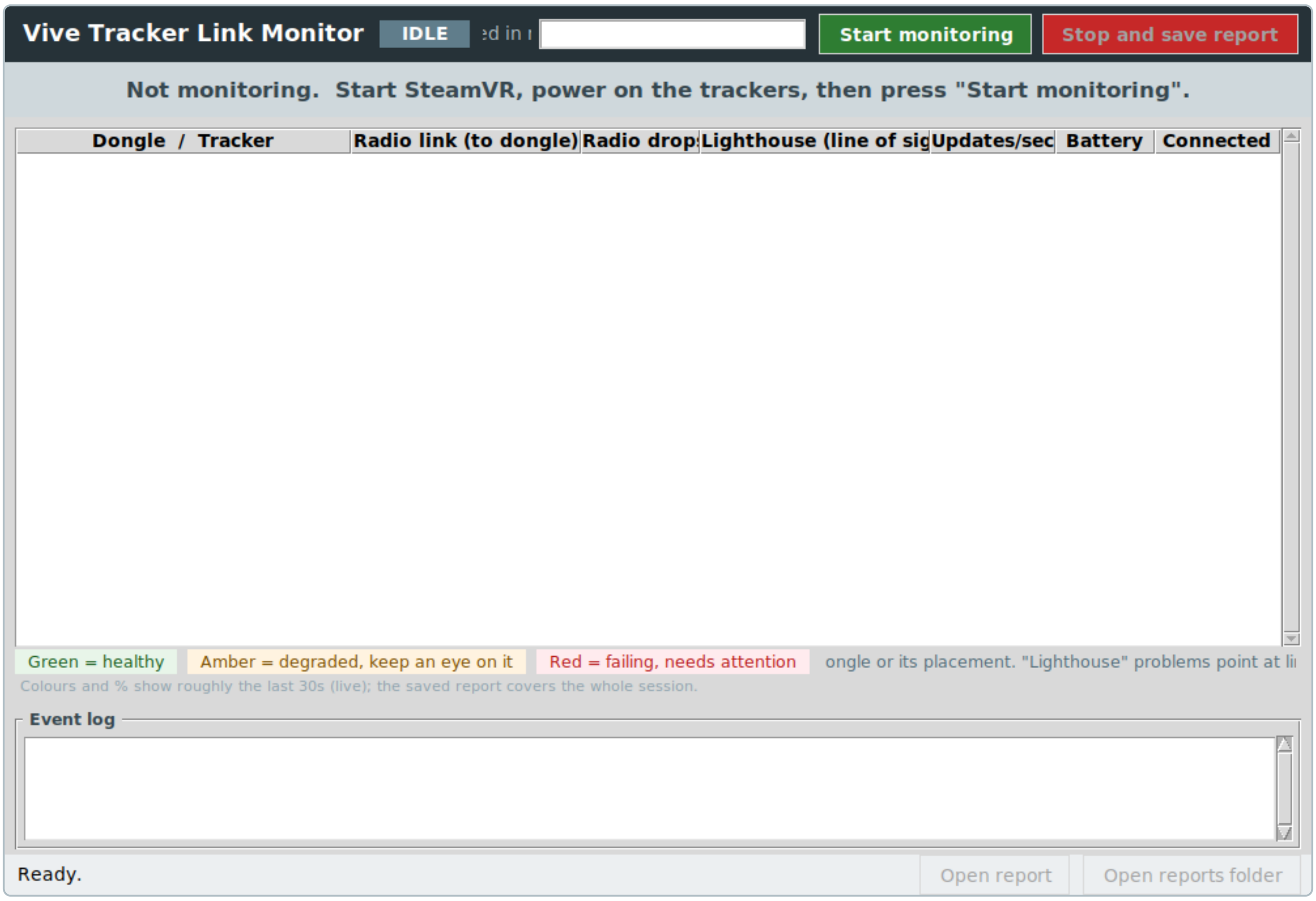
3. Run a session

Step 1 — Launch and enter a location label

Double-click `vive_dongle_gui.exe`. You'll see the idle screen.



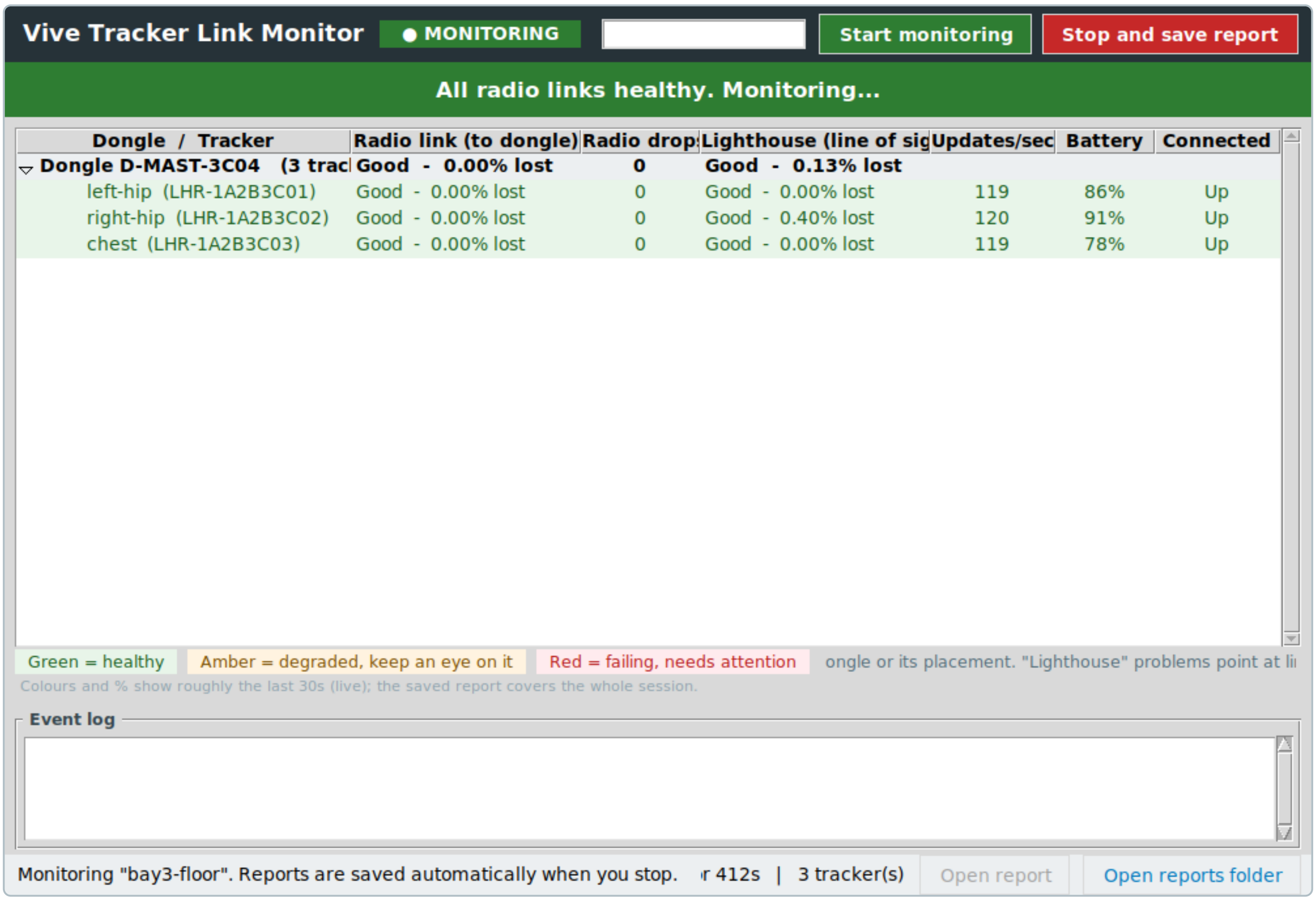
the start screen.

1. Type a **location label** in the box at the top (for example `bay3-floor`). It's required, and it's used to name the report files so different runs and locations can be told apart and compared.
2. Click **Start monitoring** (green, top right).

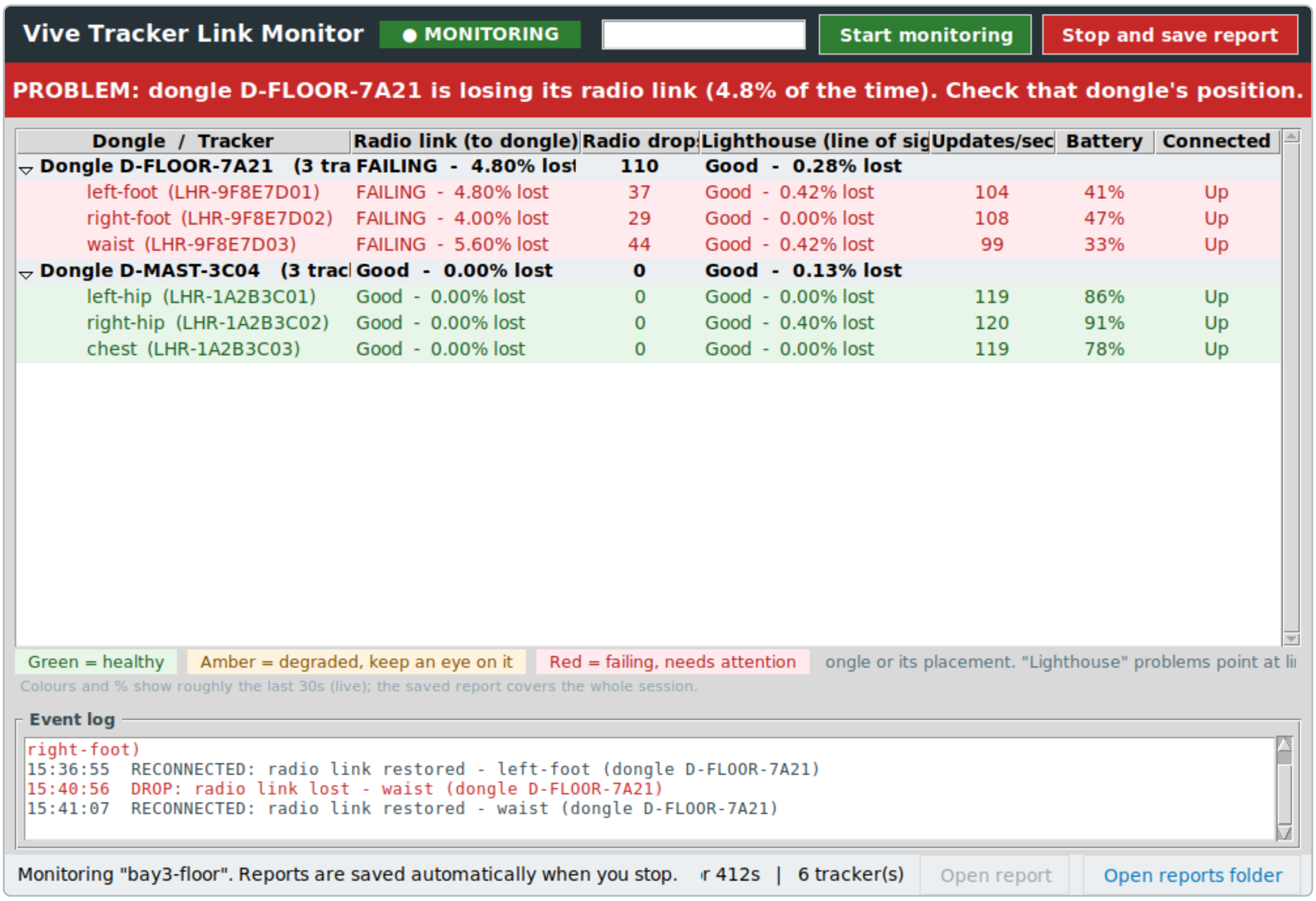
The state chip turns from `IDLE` → `CONNECTING` → `● MONITORING` once it has attached to SteamVR. If SteamVR isn't up yet, the app retries for a few seconds and then shows a dialog telling you exactly what to fix.

Step 2 — Read the live view

Every tracker appears grouped under the **dongle it is paired with**, with two verdicts side by side: **Radio link (to dongle)** and **Lighthouse (line of sight)**, each showing the percentage of time lost, plus radio drops, update rate, battery and current link state.



everything healthy; banner is green.



a failing dongle. The trackers on one dongle are red with many drops while the others stay green; the banner names the failing dongle and the Event log timestamps every drop and reconnect.

The colour, and which **side** it's on, tells you where to look:

Colour	Meaning
● Green	Healthy
● Amber	Degraded — keep an eye on it
● Red	Failing — needs attention

- A problem in the **Radio link** column → the **dongle** or its placement.
- A problem in the **Lighthouse** column → **line of sight** to the base stations.

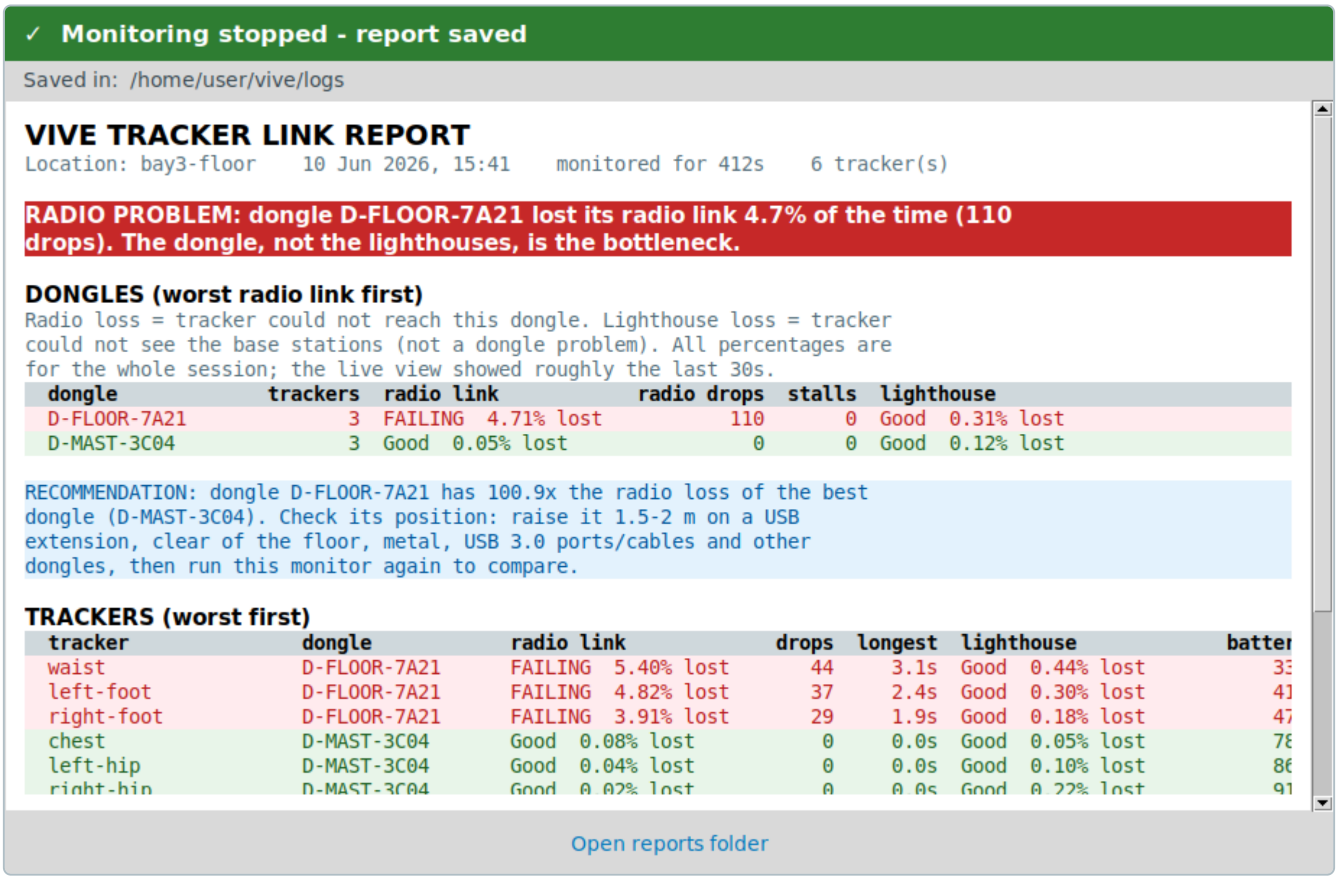
Let the session run long enough to be representative — move the rig through the poses and the area you actually care about — before stopping.

> **Live figures are "right now", not the whole run.** The colours and the "% lost" in the table reflect roughly the **last 30 seconds**, so a tracker moved back into range clears from red to green within ~30 s instead of staying flagged all session. The **saved report** uses the **full-session** figures, which is what you want for comparing runs.

Step 3 — Stop and save the report

Click **Stop and save report** (red, top right). The app:

1. Confirms on screen (green banner + a pop-up) that the report was saved, and
2. Opens a **summary window** with the full conclusion.



✓ Monitoring stopped - report saved

Saved in: /home/user/vive/logs

VIVE TRACKER LINK REPORT

Location: bay3-floor   10 Jun 2026, 15:41   monitored for 412s   6 tracker(s)

RADIO PROBLEM: dongle D-FLOOR-7A21 lost its radio link 4.7% of the time (110 drops). The dongle, not the lighthouses, is the bottleneck.

DONGLES (worst radio link first)

Radio loss = tracker could not reach this dongle. Lighthouse loss = tracker could not see the base stations (not a dongle problem). All percentages are for the whole session; the live view showed roughly the last 30s.

dongle	trackers	radio link	radio drops	stalls	lighthouse
D-FLOOR-7A21	3	FAILING 4.71% lost	110	0	Good 0.31% lost
D-MAST-3C04	3	Good 0.05% lost	0	0	Good 0.12% lost

RECOMMENDATION: dongle D-FLOOR-7A21 has 100.9x the radio loss of the best dongle (D-MAST-3C04). Check its position: raise it 1.5-2 m on a USB extension, clear of the floor, metal, USB 3.0 ports/cables and other dongles, then run this monitor again to compare.

TRACKERS (worst first)

tracker	dongle	radio link	drops	longest	lighthouse	batter
waist	D-FLOOR-7A21	FAILING 5.40% lost	44	3.1s	Good 0.44% lost	33
left-foot	D-FLOOR-7A21	FAILING 4.82% lost	37	2.4s	Good 0.30% lost	41
right-foot	D-FLOOR-7A21	FAILING 3.91% lost	29	1.9s	Good 0.18% lost	47
chest	D-MAST-3C04	Good 0.08% lost	0	0.0s	Good 0.05% lost	78
left-hip	D-MAST-3C04	Good 0.04% lost	0	0.0s	Good 0.10% lost	86
right-hip	D-MAST-3C04	Good 0.02% lost	0	0.0s	Good 0.22% lost	91

Open reports folder

the run summary.

The summary leads with a colour-coded verdict, ranks the **dongles worst radio link first**, and — when one dongle clearly stands out — gives a concrete **recommendation** (for example: raise that dongle 1.5–2 m on a USB extension, clear of the floor, metal and USB 3.0 ports, then re-run and compare).

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4. Two features worth knowing

Hide devices that aren't part of the run

Some devices show up in SteamVR but aren't what you're measuring — most often **controllers used only for room setup** and then switched off, which would otherwise sit in the table as **DOWN** for the whole session and skew the numbers.

Right-click the device's row → "Hide". A hidden device is:

- removed from the table,
- excluded from the live stats, the per-dongle aggregates, the per-second CSV **and the saved report**, and
- ignored by the "not in use" pause check (below), so a powered-off controller can't trip it.

A **"Hidden devices: N — click to show them again"** link appears under the colour legend to bring them all back. Hiding applies to the current session.

"Not in use" is not a fault

If the **headset is switched off** *or* **all (non-hidden) trackers are powered down**, the app treats it as the operator stopping — not a radio fault. The state chip shows `PAUSED (not in use)`, the banner goes grey, and that idle time is **not counted** against the loss figures and raises **no drop alarms**. It resumes automatically the moment the headset goes back on or a tracker powers up. (A brief grace period ignores momentary blips, but counting stops immediately so the pause never pollutes the numbers.)

This is why the recommended workflow is: start the run with everything on, do your room setup, switch the setup controllers off and **Hide** them, then carry on — the report will reflect only the trackers you care about, for only the time they were actually in use.

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5. Naming your trackers (optional)

Trackers are identified by serial (e.g. `LHR-9F8E7D01`). To show friendly names instead, the tool writes a plain-text **`tracker_names.txt`** next to the executable the first time it sees your trackers.

It's a normal text file — **double-click to open it in Notepad** (no Excel and no Microsoft login). Put one device per line as `SERIAL = friendly name`:

```
# Tracker names - edit this in Notepad and save.
# One device per line:   SERIAL = friendly name
LHR-9F8E7D01 = left-foot
LHR-2A41F0AB = right-foot
```

Save it, and the names appear in the table, the report and the CSV on the next run. Lines starting with `#` are ignored, and `,` , tab or `:` work as separators too. (An older `tracker_names.csv`, if present, is still read.)

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6. Where the files go

Everything is written to a **`logs\`** folder created next to the executable. Use the **Open report** and **Open reports folder** buttons in the status bar.

File	What it is
`<label>_<timestamp>_report.txt`	The same content as the summary window, in plain text.
`<label>_<timestamp>_series.csv`	One row per tracker per second — for Excel, pandas, Grafana, etc.
`<label>_<timestamp>_events.log`	Every radio drop and reconnect with a full timestamp.

CSV columns

`*_1s` columns are values for that one-second interval (what time-series tools want); `*_total` columns are cumulative since the run started.

Column	Meaning
`timestamp`, `epoch_s`	ISO time and Unix epoch for the row.
`site`	The location label you entered.
`tracker`, `tracker_name`	Serial and the friendly name (if set).
`model`, `dongle`	Tracker model and the paired dongle ID.
`connected`	1 if the radio link was up at sample time, else 0.
`rf_down_pct_1s`	% of this second the radio link was down.
`optical_oor_pct_1s`	% of this second connected-but-out-of-range.
`drops_1s`, `stalls_1s`	Radio drops and packet stalls in this second.
`rf_loss_pct_total`	Cumulative radio loss % for the run.
`optical_loss_pct_total`	Cumulative optical (out-of-range) loss % for the run.
`drops_total`, `stalls_total`	Cumulative radio drops and stalls.
`longest_disconnect_s`	Longest single radio dropout so far.
`update_hz`	Recent update rate for the tracker.
`battery_pct`	Battery level, if reported.

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7. Acting on the results

- **One dongle far worse than the rest** → a placement problem on that dongle. Raise it on a USB extension, **1.5–2 m up**, away from the floor, metal, and USB 3.0 ports/cables, then run again and compare the radio-loss percentages.
- **Radio healthy but lighthouse red/amber** → not a dongle issue. Check **line of sight** to the base stations (occlusion, range, reflective surfaces).
- **Everything green** → the links were solid for that session. Keep the saved CSV as your baseline for the next comparison.

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8. Quick troubleshooting

Symptom	What to do
"Start monitoring" then an error dialog	SteamVR isn't running/active. Start SteamVR (headset detected or Null driver) and try again.
A tracker sits as **DOWN** all session	If it's not part of the run (e.g. a setup controller), right-click → **Hide**. Otherwise it's a genuine radio drop — check that dongle.
Windows blocked the `.exe`	Right-click → Properties → **Unblock**, or disable Smart App Control briefly.
`tracker_names.txt` opens in Excel	It shouldn't — it's a `.txt`. If your PC is set to open `.txt` in Excel, right-click → Open with → **Notepad**.
Everything reads as paused / "not in use"	The headset is off or all trackers are powered down. Put the headset on or power a tracker; it resumes automatically.